Information agents for the World Wide Web

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This paper describes a distributed system of intelligent agents, Jasper, for performing information tasks over the Internet World Wide Web (WWW) on behalf of a community of users. Jasper can summarise and extract keywords from WWW pages and can share information among users with similar interests automatically. Jasper provides agents which can retrieve relevant WWW pages quickly and easily. A Jasper agent holds a profile of its user, based on observing their behaviour and learning more about their interests as the system is used.

A novel three-dimensional front end on to the Jasper system has been created using VRML (virtual reality modelling language), a language for 3-D graphical spaces or virtual worlds networked via the global Internet and hyperlinked within WWW. This and other ongoing research using keyword and document clustering techniques are described.

1. Introduction

In 1982, the volume of scientific, corporate and technical information was doubling every five years. By 1988, it was doubling every 2.2 years and by 1992 every 1.6 years. With the expansion of the Internet and other networks this rate of increase will continue. Key to the viability of such networks will be the ability to manage the information and provide users with what they want, when they want it.

This paper discusses a distributed system of intelligent agents for performing information tasks over the Internet World Wide Web (WWW) on behalf of a user or community of users, describing how agents are used to store, retrieve, summarise and inform other agents about information found on WWW in a system called Jasper (joint access to stored pages with easy retrieval). In certain circumstances, Jasper agents will also identify an opportunity for performance improvement — by observing its user, an agent will attempt to enhance the profile it holds on that user.

The approach here is not motivated by any perceived requirement for another tool for searching WWW — there are already many of these [1, 2] and they are being added to frequently with ever-increasing coverage of the Web and sophistication of search engines. The motivation behind this project is different but related — having found useful information on WWW, how can it be stored for easy retrieval and how can other users, likely to be interested in the information, be identified and informed?

Given the vast amount of information available on WWW, it is preferable to avoid the copying of information from its original location to a local server. Indeed, it could be argued that approach is contrary to the whole ethos of the Web. Rather than copying information, therefore, Jasper agents store only relevant meta-information. As will be shown below, this includes keywords, a summary, document title, universal resource locator (URL) and date and time of access. This meta-information is then used to index on the actual information when a retrieval request is made.

Most current WWW clients (Mosaic, Netscape, and so on) provide some means of storing pages of interest to the user. Typically, this is done by allowing the user to create a hierarchical menu of names associated with particular URLs. While this menu facility is useful, it quickly becomes unwieldy when a reasonably large number of WWW pages are involved. Essentially, the representation provided is not sufficiently rich to permit a useful index of the information stored — the user can only provide a string naming the page. As well as the fact that useful meta-information such as the date of access of the page is lost, a single phrase (the name) may not be enough to accurately index a page in all contexts. Consider, as a simple example, information about the use of knowledge-based systems (KBS) in information retrieval of pharmacological data — in different contexts, it may be KBS, information retrieval or pharmacology which are of interest. Unless a name is carefully chosen to mention all three aspects, the information will be missed in one or more of its useful contexts. This problem is analogous to the problem of finding files containing desired information in a Unix (or other) file system [3], although in most filing systems one at least has the facility to sort files by creation date.

The solution adopted for this problem was to allow the user to access information by a much richer set of meta-information. How Jasper agents achieve this and how the resulting meta-information is exploited is the subject of the next section.

2. Agent architecture

This section discusses the facilities which Jasper agents offer the user in managing information. These can be grouped in two categories — storage and retrieval.

2.1 Storage

Figure 1 shows the actions taken when Jasper stores information in its page store (JPS). The user first finds a WWW page of sufficient interest to be stored by Jasper in their JPS and sends a 'store' request to Jasper via a menu option on their favourite WWW client (Mosaic and Netscape versions are currently available on all platforms). Jasper then invites the user to supply an annotation to be stored with the page. Typically, this might be the reason the page was stored and can be very useful for other users in deciding which pages retrieved from JPS to visit. Information sharing is discussed further below.

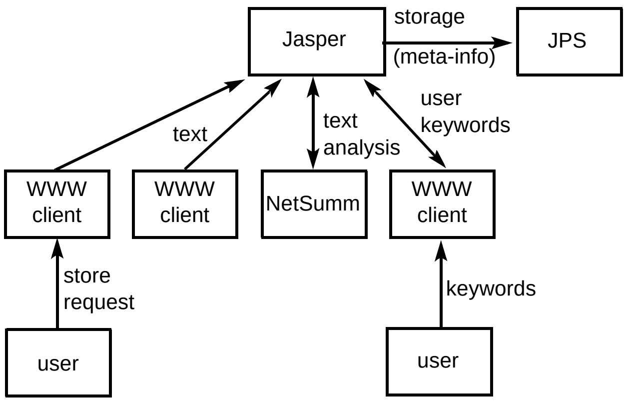

Fig 1 Jasper's storage process.

Jasper next extracts the source text from the page in question, first stripping out html (Hypertext Mark-up Language) tags. Jasper then summarises the text using NetSumm, BT's automatic text summariser [4]. Jasper also extracts and counts the words occurring in the text, first filtering out a standard 'stop-list' of commonly occurring terms.

At the end of this process, Jasper has the following meta-information about the WWW page of interest:

- a set of keywords (indexing terms),
- the user's annotations,
- a summary of the page's content,
- the document title,
- universal resource locator (URL),
- date and time of storage.

Jasper then adds the page to the JPS. In the JPS, the keywords (of both types) are then used to index on files containing the other meta-information, as shown in Fig 2.

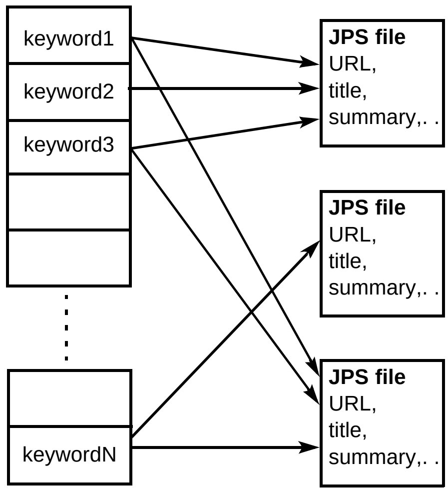

Fig 2 JPS Structure.

2.2 Retrieval

There are three modes in which information can be retrieved from JPS using Jasper. One is a standard keyword retrieval facility, while the other two are concerned with information sharing between a community of agents and their users. Each is described below.

When a Jasper agent is installed on a user's machine, the user provides a personal profile — a set of keywords which describe information the user is interested in obtaining via WWW. This profile is held by the agent in order to determine which pages are potentially of interest to its user.

Keyword Retrieval

As is shown in Fig 3, for straightforward keyword retrieval the user supplies a set of keywords to their Jasper agent via an html form provided by Jasper. The Jasper agent then retrieves the ten most closely matching pages held in JPS, using a simple keyword matching and scoring

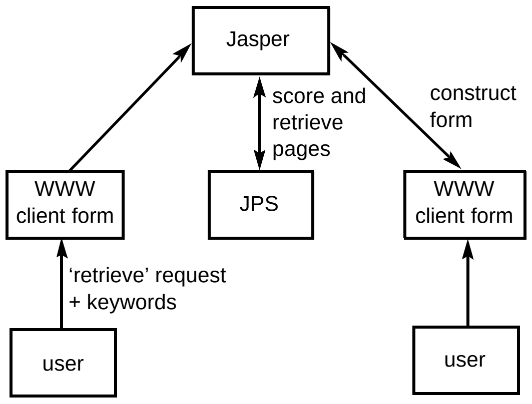

Fig 3 Jasper's keyword retrieval process

algorithm. The user can specify in advance a retrieval threshold below which pages will not be displayed. The agent then dynamically constructs an html page with a ranked list of links to the pages retrieved and their summaries. Any annotation made by the original user is also shown, along with the scores of each retrieved page. This page is then presented to the user on their WWW client.

'What's new?' facility

Any user can ask his Jasper agent 'What's new?' The agent then interrogates the JPS and retrieves the most recently stored pages. It then determines which of these pages best match the user's profile based on the same keyword matching algorithm as that mentioned above. An html page is then presented to the user showing a ranked list of links to the recently stored pages which best match the user's profile and the other pages most recently stored in JPS, with annotations where provided. In this way users are provided with a view both of the pages recently stored of most interest to themselves and of a more general selection of recently stored pages.

A user can update the profile which their Jasper agent holds at any time via an html form which allows the addition and/or deletion of keywords from the profile. In this way, the user can effectively select different contexts in which to work. A context is defined by a set of keywords (those making up the profile or, indeed, those specified in a retrieval query) and can be thought of as those types of information in which a user is interested at a given time.

The idea of applying human memory models to the filing of information was explored by Jones [3] in the context of computer filing systems. As he pointed out in the context of a conventional filing system, there is an analogy between a directory in a file system and a set of pages retrieved by a Jasper agent. The set of pages can be thought of as a dynamically-constructed directory, defined by the context in which it was retrieved. This is a highly flexible notion of 'directory' in two senses — firstly, pages which occur in this retrieval can of course occur in others, depending on the context, and, secondly, there is no sharp boundary to the directory, since pages are 'in' the directory to a greater or lesser extent depending on their match to the current context. In this approach, the number of ways of partitioning the information on the pages is thus only limited by the diversity and richness of the information itself.

Communication with other interested agents

Jasper agents are currently being tested by a group of users on a WWW server at BT Laboratories. When a page is stored in JPS by a Jasper agent, the agent checks the profiles of other agents' users in its 'local community' (here the agents in the trial, although this could be any predefined community). If the page matches a user's profile with a score above a certain threshold, an e-mail message is automatically generated by the agent and sent to the user concerned, informing him of the discovery of the page. The e-mail header is in the format:

JASPER KW: keyword $_1$ ... keyword $_n$

This allows the user (before reading the body of the message) to identify it as being one from Jasper and since a list of keywords is provided the user can assess the relative importance of the information to which the message refers. The keywords in the message header vary from user to user, depending on the keywords from the page which match the keywords in their user profile, thus personalising the message to each user's interests. The message body itself informs the user of the page title and URL, who stored the page, and any annotation to the page which the storer provided. Other forms of automatic notification (e.g. via dynamically constructed WWW pages) are also being studied.

As argued in section 4 below, the ability to share relevant WWW information with other users quickly and easily is a key part of the original vision of WWW. A Jasper agent can automate this information task for a user to some extent.

2.3 Learning from feedback

In certain circumstances, Jasper agents will identify an opportunity for performance improvement and will seek user feedback in order to achieve the desired performance enhancement.

Jasper agents can modify users' profiles if a Jasper agent identifies that the profile does not reflect the pages which the user is storing. When a page is stored (see section 2.1), the agent compares the content of the page with the user's profile. If the profile does not match the page content above a given threshold, Jasper informs the user and invites the user to modify their profile by adding some keywords suggested by the agent based on the page's content. The user can then add these keywords to the profile or add some new keywords of their own to the profile.

3. Clustering in Jasper

It has been shown that the Jasper store is essentially what would be called a 'collection' in information retrieval (IR) terminology — it is a set of documents indexed by keywords. It differs from a 'traditional' collection in that the documents are typically located remotely from the index — the index actually points to a URL which specifies the location of the document on the Internet. Furthermore, various additional pieces of meta-information are attached to documents in Jasper, such as the user who stored the page, when it was stored, any annotation the user may have provided, and so on.

One additional important area where Jasper differs from most document collections is that all documents in the Jasper store have been entered by users who have made a conscious decision to mark it as a piece of information which they and their peers would be likely to find useful in the future. This, along with the meta-information held, makes a Jasper store a very rich source of information.

Given the similarity between a Jasper store and a standard document collection mentioned earlier, it is sensible to examine whether IR techniques can be applied beneficially to the Jasper store. Accordingly, the use in Jasper of one such technique, clustering, has recently been under investigation.

3.1 Clustering documents

Jasper's page store (see Fig 2) can be viewed as an example of a term-document matrix, where each element of the matrix m_{ij} represents the number of times term (keyword) j occurs in document (page) i. Jasper's term-document matrix is used to calculate a similarity matrix for the documents in the Jasper store. The similarity matrix gives a measure of the similarity of documents in the store. For each pair of documents, the 'Dice coefficient' is calculated. For two documents D_i and D_j, the Dice coefficient is given by:

$$2 \times |D_i \cap D_j| / |D_i| + |D_j|$$

where $|D_i|$ is the number of terms in document D_i and $|D_i \cap D_j|$ is the number of terms co-occurring in documents D_i and D_j. This coefficient yields a number between 0 and 1. A coefficient of zero implies two documents have no terms in common, while a coefficient of 1 implies that the sets of terms occurring in each document are identical. The similarity matrix, Sim say, represents the similarity of each pair of documents in the store as calculated by the Dice coefficient, so that for each pair of documents i and j:

$$\mathrm{Sim}(i,j) = 2 \times |D_i \cap D_j| / |D_i| + |D_j|$$

This matrix is used to create clusters of related documents automatically using the hierarchical agglomerative clustering process [5—7], whereby each document is initially placed in a cluster by itself and the two most similar such clusters are then combined into a larger cluster, for which similarities with each of the other clusters must then be computed. This combination process is continued until only a single cluster of documents remains at the highest level. The way in which similarity between clusters (as opposed to individual documents) is calculated can be varied. The details are of no concern here other than to say that for the Jasper store complete-link clustering has been employed. In complete-link clustering, the similarity between the least similar pair of documents from the two clusters is used as the cluster similarity.

3-D interfaces to document collections

The resulting cluster structures of the Jasper store are being used to create a novel three-dimensional (3-D) front-end on to the Jasper system, using VRML (virtual reality modelling language) [8]. VRML is a language for 3-D graphical spaces or virtual worlds networked via the global Internet and hyperlinked within the World Wide Web (WWW).

A major consequence of the introduction of WWW to the Internet has been an increase in the usability of the information within it. This is directly attributable to the navigability of the information — in other words, the Internet is useful (and will be used) to the degree it is capable of conforming to requests made of it. Furthermore, it has added a universal structure to the information within it; through WWW, all four million Internet hosts can be treated as a single, unified data source, and all of the information can be treated as a single, albeit highly complex, document.

Navigability in a purely symbolic domain has limits. The amount of 'depth' present in a subject before it exceeds human capacity for comprehension (and hence, navigation) is finite and relatively limited. However, humans have a sophisticated visual system and the ability to navigate in three dimensions. It thus seems reasonable to propose that the WWW should be extended, increasing its navigation model from two to three dimensions.

It is proposed to exploit these insights within the Jasper system by constructing a shared 3-D space, dubbed the 'Information Garden', through and over which users can fly and wherein information is arranged in a logical, intuitive and accessible way. This arrangement of information is achieved by using the meta-information in the Jasper store and results from the clustering process (described above). Figure 4 shows an early version of the Information Garden. It should be stressed that the current interface is a tentative early design. The Information Garden is a VRML 3-D representation of the Jasper store with documents clustered into various groups. However, it is also viewed as a shared virtual space, in which users (represented by Avatars) can meet, communicate and exchange information.

Figure 4 is a screenshot from a working prototype shared space where avatars can meet, access information via the Information Garden, and communicate via a number of media including text, audio and video. Shared electronic whiteboards are also provided. As the user navigates towards a particular sector, more detail becomes visible, as shown in Fig 5.

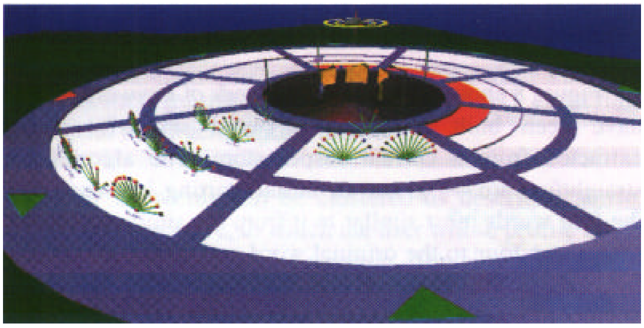

Fig 4 Shared information space.

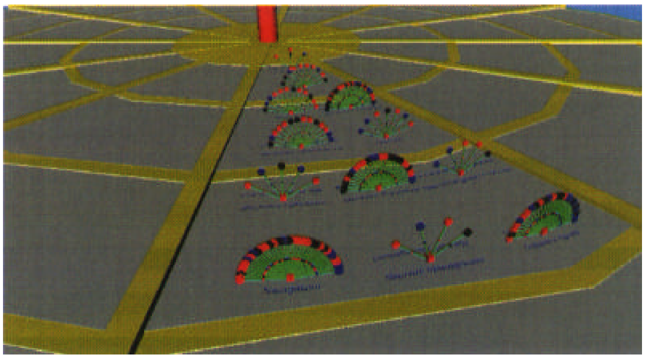

Fig 5 Information garden — view 1.

In Fig 5 can be seen clustered groups of documents. Each document is represented by a stalk with a coloured circle at its end. Figure 6 shows a smaller part of the same garden in more detail. The colour coding of the circles on the stalks represents the status of the particular document — red circles indicate that a document has been updated since it was stored in Jasper, black that the link is 'dead' (that is, the URL associated with this document is no longer valid), while a blue circle indicates that neither of these conditions apply. The smaller circles half-way up the document stalk represent a locally held summary of the information, as generated by Jasper.

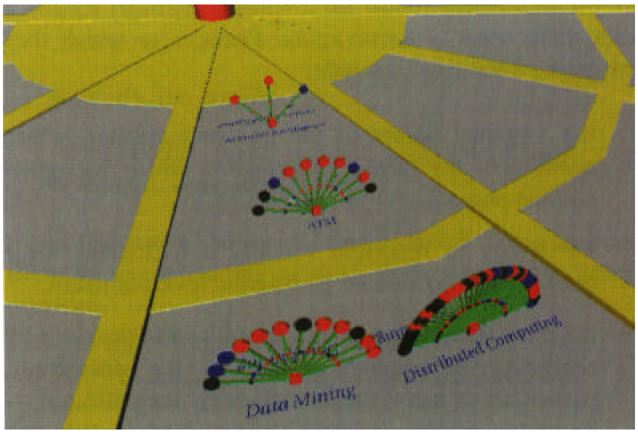

Fig 6 Information garden — view 2.

At some point, it makes sense to drop down from a 3-D to a 2-D view of the information and by clicking on the large or small circles, the user is shown a WWW page of the original information or the summary respectively. Alternatively, the user can click on the base of the cluster. The system will then present the user with a view of a single cluster, as shown in Fig 7. At this viewpoint, each stalk of the cluster is annotated with the title of the document it represents. As before, the document or its summary can be accessed from here by clicking on the relevant circle.

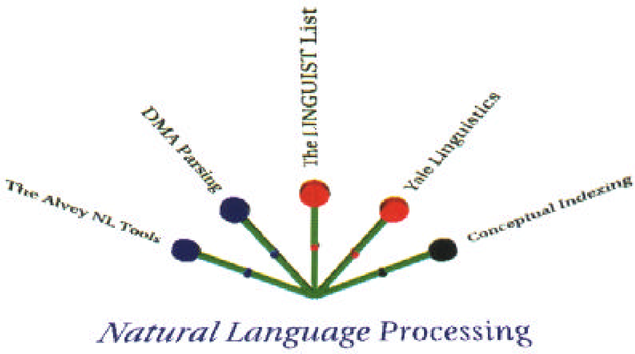

Fig 7 Information garden document cluster.

As mentioned above, the preceding paragraphs describe 'work in progress'. A number of issues remain to be resolved, including how to generate clusters of reasonable size for this type of representation and how best to label such clusters. One possibility is to label a cluster with the most commonly occurring terms [9]. Furthermore, it may prove that other metaphors are more appropriate for visualisation of information.

Evaluation of such an interface is also a key issue which has not yet been addressed to any great extent. The shared space is currently being used by a few expert users and is not yet sufficiently robust for a wider trial. Early feedback from those users is encouraging. Features to which they have reacted positively include:

- the ‘organic’ nature of the garden metaphor, which encourages the idea of information which will grow (and be pruned),
- the ability to represent visually a reasonably large collection of documents at different levels of detail,
- the meta-information about the collection provided by clustering and colour coding of the documents, particularly when viewed at relatively long distance — examples of the information provided include the amount of information available on a given topic and clusters where many links have ‘died’ or need updating,
- the tendency to get ‘lost in hyperspace’ is reduced, since the user can stay in the same 3-D space and yet have access to a range of information and meta-information.

Work is continuing in experimenting with this and other metaphors. One current investigation involves the use of spatial information — for example, given that Jasper holds a profile of a user, the document clusters shown in the Information Garden could be spatially arranged such that those likely to be of most interest are closest to the user’s entry point to the 3-D space.

Clustering keywords

Keywords (terms) occurring in a particular collection can also be clustered in a way which mirrors exactly the document clustering technique described above: a similarity matrix for the keywords in the Jasper store can be constructed which gives a measure of the ‘similarity’ of keywords in the store. For each pair of documents, the Dice coefficient is calculated. Using the same notation as for document clustering, for two keywords K_i and K_j, the Dice coefficient is given by:

$$2 \times |K_i \cap K_j| / |K_i| + |K_j|$$

Once the similarity matrix for a Jasper store is calculated, however, the keywords are not clustered as the documents were, but instead the matrix itself is exploited in two ways.

The first way is profile enhancement. A user’s profile is enhanced by adding those keywords most similar to the keywords explicitly represented in the user’s profile in a way reminiscent of query reformulation techniques, such as the use of spreading activation networks [10, 11] and other techniques based on the use of semantic nets [12, 13].

In Chakravarthy and Haase [12], for example, the use of semantic knowledge from WordNet [14] in a program called NetSerf for finding information archives on the Internet is reported. Based on a set of 75 queries, NetSerf was reported to have performed significantly better than the Smart [15] information retrieval system. Interestingly, local experiments with WordNet for profile enhancement have found that WordNet did not perform well with respect to the very technical vocabulary which constitutes the terms in the Jasper store. It was this technical bias that led to looking to the store itself for a network of related terms — in effect, the clustered terms constitute a simple spreading activation network. In Wettler and Rapp [11], some results from cognitive psychology research are analysed to explain why it is difficult for people to think of related terms. This difficulty is ameliorated to a degree by the automatic (hidden) enhancement of a user’s profile by the system.

Figure 8 shows an example network of keywords which have been built from the keyword similarity matrix extracted from a current Jasper store. The algorithm is straightforward — given an initial starting keyword, find the four words most similar to it from the similarity matrix. Link these four to the original word and repeat the process for each of the four new words. This can be repeated a number of times (in Fig 8, three times). One could of course attach the particular similarity coefficients to each link for finer-grained information concerning the degree of similarity between words.

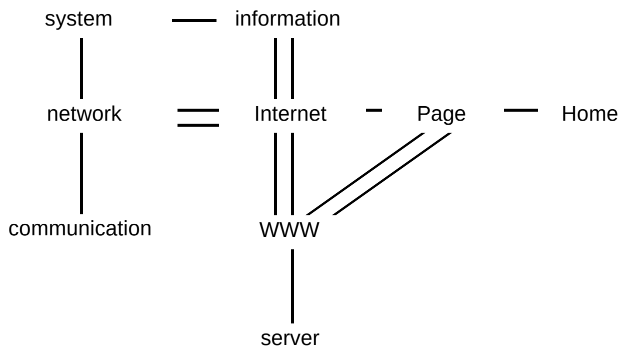

Fig 8 Clustered keywords from the Jasper store.

Thus, for example, if the words ‘virtual’, ‘reality’ and ‘Internet’ are part of a user’s profile but ‘VRML’ is not, an enhanced profile might add VRML to the original profile (assuming VRML is clustered close to ‘virtual’, ‘reality’ and ‘Internet’). In this way, documents containing ‘VRML’ but not ‘virtual’, ‘reality’ and ‘Internet’ may be retrieved whereas they would not have been with the unenhanced profile.

The second way keyword clustering is exploited is proactive searching. The keywords comprising a user’s profile can be used by Jasper to search proactively for new WWW pages relevant to their interests, which can then present a list of new pages in which the user may be interested without them having to explicitly carry out a search. These proactive searches can be carried out by Jasper at some given interval, perhaps weekly or fortnightly. Clustering is useful here because a profile may reflect more than one interest. Consider, for example, the following user profile:

{internet, WWW, html, http, football, Manchester, United, linguistics, parsing, pragmatics}

Clearly, three separate interests are represented in the above profile and searching on each separately is likely to yield far superior results than merely entering the whole profile as a query for the given user. It is hoped that clustering keywords from the document collection will automate the process of query generation for proactive searching by a user's Jasper agent. A typical profile is:

{internet, information, retrieval, SMART, clustering, agent, intelligent, corba, idl, dce}

These keywords can be clustered as described above, using the Jasper store, to which the user whose profile this is belongs, as the document collection. The similarity matrix shown in Table 1 is then obtained (the upper right half of the matrix is of course a mirror image of the bottom left). If complete-link clustering is used [5, 7], whereby the similarity between the least similar pair of items from two clusters is taken as the similarity between the clusters, the cluster dendogram shown in Fig 9 is obtained. Note that the vertical axis in Fig 9 represents difference rather than similarity. If a similarity threshold of 0.3 is set, beneath which any clusters are disregarded, the following clusters are obtained:

{dce corba idl}
{SMART clustering}
{internet information}
{intelligent agent}
{retrieval}

Similarly, the profile:

{java, corba, agent, internet, information, golf, vrml, realaudio, vrml, http, html}

yields the matrix shown in Table 2, which, when clustered using the complete-link method, gives the clusters shown overleaf:

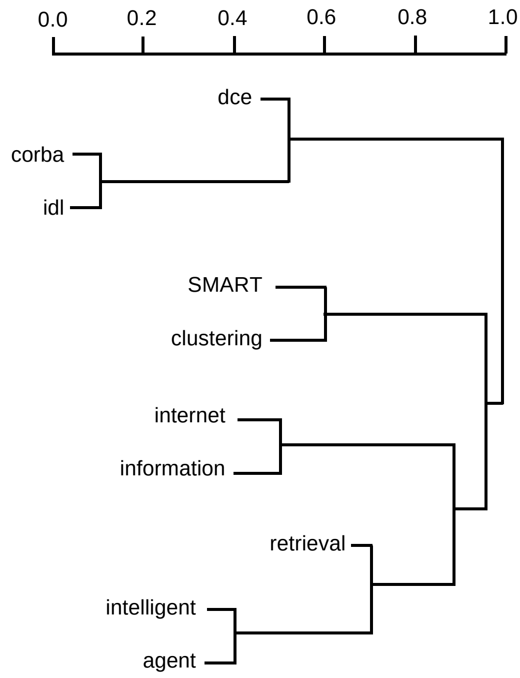

Fig 9 Dendogram 1.

Table 1 Term similarity — Matrix 1.

	corba	dce	idl	internet	information	retrieval	intelligent	agent	SMART	clustering
corba	1									
dce	0.62	1								
idl	0.92	0.50	1							
internet	0.03	0.03	0.03	1						
information	0.04	0.04	0.03	0.51	1					
retrieval	0.04	0.08	0.00	0.21	0.27	1				
intelligent	0.00	0.03	0.00	0.21	0.30	0.30	1			
agent	0.07	0.11	0.04	0.16	0.29	0.38	0.53	1		
SMART	0.00	0.00	0.00	0.07	0.07	0.19	0.10	0.14	1	
clustering	0.00	0.00	0.00	0.01	0.03	0.17	0.04	0.12	0.43	1

{internet information}
{http html}
{agent}
{java}
{corba}
{golf}
{realaudio}
{vrml}

Inspection of the results of clustering these and other profiles indicates that clustering is able to subdivide a Jasper profile into sets of keywords which may prove valid for use as queries. More work remains to be done, however, to determine the best clustering technique (single-link, complete-link or group average) and the most appropriate clustering threshold for a wider range of profiles. It may prove necessary to attempt to tune dynamically the threshold (and indeed perhaps the clustering technique) based on the similarity matrix for a given profile. Consideration is also being given to a semi-automatic technique, whereby the set of queries created by Jasper from the user's profile is presented to the user for confirmation or adjustment before any queries are actually submitted.

Because queries based on profile clusters are made off-line by Jasper, rather than interactively by the user, there is an opportunity to analyse the results before presenting them to the user. For example, when the search results are obtained by Jasper, they can be summarised and matched against the user's profile in the usual way to give a prioritised list of new URLs along with locally held summaries. In addition, a log can be held of the pages which Jasper has found for each user. By checking against this log, only new pages (those the user has not been shown by Jasper previously) will be presented. It is anticipated that the pages will be presented to the user on a dynamically constructed WWW page with the facility to store pages of particular interest in the Jasper store easily. Those pages accepted for storage in Jasper by this mechanism (and indeed those rejected) may additionally afford new opportunities for user-profile adaptation.

4. Related work

Jasper is an example of a static information agent — its goal is to help the user manage the 'information overload' problem often encountered when using the WWW. However, it does not traverse the WWW on behalf of its user but rather acts as an aid to the user at the user's point of interaction with the WWW. In this sense, it is an interface agent in the sense of Maes [16] — it is acting as an assistant to reduce the information overload of a user (by supplying good quality WWW pages as pre-filtered by other users), and, by observing user actions, it will adapt its own behaviour to better reflect its user's needs (by modifying users' profiles based on the pages they store).

The services provided by existing Internet information-retrieval tools can be divided into four main functions — search, storage, access and organisation.

There are many systems which offer some or all of these to the WWW user, including WAIS [17], Archie [18] and the Harvest system [19]. Harvest is one of the most sophisticated of these types of system and it is briefly described here, and compared with Jasper. All the systems mentioned above can be characterised as 'off-line' systems in the sense that their indexes and stores are not built incrementally, as in Jasper, but are rather constructed off-line for later use. The key elements of the Harvest approach are gatherers and brokers. Gatherers collect indexing data from a given collection of information (http, ftp and gopher protocols are supported). Brokers provide query interfaces to the gathered information. Brokers can access more than one gatherer, as well as other brokers. Experiments indicate that Harvest reduces load on servers and networks. This is due to efficient gathering software and the sharing of

Table 2 Term similarity — Matrix 2.

	http	html	java	corba	realaudio	vrml	internet	golf	agent	information
http	1									
html	0.044	1								
java	0.10	0.21	1							
corba	0.30	0.06	0.10	1						
realaudio	0.00	0.00	0.00	0.00	1					
vrml	0.13	0.12	0.07	0.00	0.00	1				
internet	0.16	0.26	0.20	0.03	0.03	0.09	1			
golf	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1		
agent	0.05	0.11	0.07	0.07	0.00	0.00	0.16	0.00	1	
information	0.14	0.26	0.12	0.04	0.01	0.09	0.51	0.00	0.29	1

information among indexes that need it, in contrast to other comparable information retrievers which use expensive object-retrieval protocols and fail to co-ordinate information gathering among themselves.

Harvest queries on WWW brokers return references to relevant information sources, an indication of the degree of match to the query and a content summary. The summarisation performed on text files is simply to extract the first 100 lines plus the first sentence of each remaining paragraph. Keywords seem to be an alphabetical list of this summary. It is possible in principle to write and plug in a personalised summariser. Harvest also 'summarises' other formats of information, albeit in a fairly simple way. The summary of an audio file, for example, is the file name, while the summary of a perl script is the procedure names and comments therein.

Harvest and Jasper thus differ in several ways — firstly, Harvest is inherently 'off-line' as discussed above while Jasper is 'incremental', and, secondly, Harvest summaries are more simplistic than those attempted by Jasper, although Harvest provides 'summaries' of a wider range of information types.

More similar to Jasper than off-line indexers of information is 'Warmlist' [20], a tool for caching, searching and sharing WWW documents. Like Jasper, Warmlist extends the idea of the hotlist. WWW documents in the Warmlist are automatically cached on the local server, along with the original links to other information. This gives much quicker access to Warmlist pages. A useful feature of a Warmlist is the ability to include other Warmlists as part of one's own Warmlist. The Glimpse indexing and searching package [21] is used to search cached pages.

In Jasper, a different approach is adopted by not storing whole pages but rather storing meta-information about a page. As described above, this allows much enhanced, richer indexing on pages of interest without the necessity of copying remote information to local servers. It is possible that with many users on a WWW server using Warmlist, the server would rapidly fill up with cached pages. Also the concept of a page being copied and stored in many different places (i.e. on multiple Warmlists) seems somewhat against the ethos of the Web. Given that Warmlist caches entire pages, it is unsurprising that it provides no facilities for information summarisation.

WWW documents in a Warmlist can be organised in a hierarchical way with nested directories. As argued above, however, a more flexible way to access information is via a set of keywords describing the contents of the page. This removes the necessity to remember where documents have been stored (e.g. 'Did I store the letter to Smith about the Internet in letters/smith/Internet or Internet/letters or smith/letters or ...?')

As mentioned above, there are four main functions (search, storage, access and organisation) common to many Internet information tools. In the Jasper system, the first steps have been taken towards adding a fifth function — automated information sharing, as described in section 2.2. It is not thought that this functionality is provided by any other Internet tool at the current time.

5. Conclusions

In his seminal article, Bush [22] describes a tool to aid the human mind in dealing with information. He states that previous scientific advances have helped humans in their interactions with the physical world but have not assisted humans in dealing with large amounts of knowledge and information. Bush proposed a tool called a 'memex' which could augment human memory through associative memory, where related pieces of information are linked. Trails through these links could then be stored and shared by others. WWW itself fulfils Bush's vision in some respects — Bush's associative memory can be seen in the hyperlinks of WWW. What is lacking is a way of organising this vast 'memory' of WWW pages into coherent 'trails' which can be saved and communicated to others. Currently, only relatively simplistic hotlists and menus are available.

Jasper goes some way to addressing these problems by providing agents which, as has been shown, can store meta-information about WWW pages which can then be used to retrieve relevant pages quickly and easily and share the information contained in those pages with other users with the same interests. Jasper leaves aside the issue of how best to search WWW for information (many other researchers are working on this) and is an attempt to address the complementary problem of how best to store information once it has been found and how to share information with others with the same interests. As described in sections 3 and 5, much remains to be done — in particular, the exploitation of more of the meta-information obtained by Jasper agents and the provision of adaptive agents which can infer context from users' actions will be a useful enhancement to the current system. However, Jasper is a step along the road towards the original vision for WWW [23] as a network which supports co-operative working and the sharing of information.

References

1 http://lycos.cs.cmu.edu/

2 http://www.infoseek.com/

3 Jones WP: 'On the applied use of human memory models: the memory extender personal filing system', Int J Man-Machine Studies, 25, pp 191—228 (1986).

4 http://www.labs.bt.com/innovate/informat/netsumm/index.htm

5 Rasmussen E: ‘Clustering Algorithms’, in Frakes W B and Baeza-Yates R (Eds): ‘Information Retrieval: Data structures and Algorithms’, Prentice-Hall, London (1992).

6 Frakes W B and Baeza-Yates R (Eds): ‘Information Retrieval: Data Structures and Algorithms’, Prentice-Hall, London (1992).

7 Griffiths A, Robinson L A and Willett P: ‘Hierarchic Agglomerative Clustering Methods for Automatic Document Classification’, J of Documentation, 40, No 3, pp 175—205 (1984).

8 Raggett D: ‘Extending WWW to support Platform Independent Virtual Reality’, http://vrml.wired.com/concepts/raggett.html

9 van Rijsbergen C J: ‘Information Retrieval’, Butterworths, London (1979).

10 Ruge G: ‘Human Memory Models and Term Association’, Proc 18th Annual International ACM SIGIR Conference, Washington, USA (1995).

11 Wettler M and Rapp R: ‘A Connectionist System to Simulate Lexical Decisions in Information Retrieval’, in Pfeifer R, Schreter Z, Fogelman F and Steels L (Eds): ‘Connectionism in Perspective’, Elsevier, Amsterdam (1989).

12 Chakravarthy A S and Haase K B: ‘NetSerf: Using Semantic Knowledge to Find Internet Information Archives’, Proc 18th Annual International ACM SIGIR Conference, Washington, USA (1995).

13 Rada R and Bicknell E: ‘Ranking Documents Based on a Thesaurus’, Journal of the American Society for Information Science, 40, No 5, pp 304—310 (1989)

14 Miller GA: ‘WordNet: An Online Lexical Database’, International Journal of Lexicography, 3, No 4, (1990).

15 Salton G: ‘The SMART Retrieval System’, Englewood Cliffs, NJ, USA, Prentice-Hall (1971).

16 Maes P: ‘Agents that Reduce Work and Information Overload’, Comm ACM, 37, No 7 (1994).

17 Brewster K and Medlar A: ‘An Information System for Corporate Users: Wide Area Information Servers’, Connections — The Interoperability Report, 5, No 11 (November 1991), also available from ftp://think.com/wais/wais-corporate-paper.text

18 Emtage A and Deutsch P: ‘Archie: an electronic directory service for the Internet’, Proc Usenix Winter Conference (January 1992).

19 Bowman C M, Danzig P B, Hardy D, Manber U and Schwartz M: ‘The Harvest Information Discovery and Access System’, Proc 2nd Intl WWW Conf, Chicago, Illinois, US (October 1994).

20 Klark P and Manber U: ‘Developing a Personal Internet Assistant’, http://glimpse.cs.arizona.edu:1994/~paul/warmlist/paper.html

21 Manber U and Wu S: ‘GLIMPSE: A Tool to Search through Entire File Systems’, Usenix Winter Technical Conference, San Fransisco, US (January 1994).

22 Bush V: ‘As We May Think’, The Atlantic Monthly (July 1945), also available as http://www.csi.uottawa.ca/~dduchier/misc/vbush/as-we-may-think.html

23 Berners-Lee T: http://www.w3.org/pub/WWW/

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